

AI for Medicine – Project Report

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Note: The report should be concise but complete. Typical length: 8–10 pages.

1. Project Title

Adversarial Robustness and Data Integrity of Neural Networks in Clinical Decision Support

“A Cost-Aware Security Audit of a Multi-Layer Perceptron for Heart Disease Prediction”

2. Problem Statement

The integration of Artificial Intelligence (AI) into preventive healthcare offers immense potential for the early detection and management of severe medical conditions. However, the deployment of these predictive models introduces critical security and reliability risks that threaten the core principles of **Trustworthy AI**. While adversarial machine learning has been heavily studied in image and text domains, vulnerabilities within tabular data—such as Electronic Health Records (EHRs)—remain significantly under-explored (**Dyrmishi et al., 2025**).

Unlike images, where adversarial noise is often imperceptible to the human eye, tabular medical data is governed by strict structural constraints and domain-specific rules (e.g., resting blood pressure must be a realistic whole number). Consequently, evaluating the true security of a medical AI requires specialized frameworks that account for both the mathematical optimization of the attack and the clinical realism of the modified data.

Historically, adversarial attacks on tabular data fail because they produce continuous, out-of-range values that violate medical feasibility and would instantly trigger hospital anomaly alarms. Therefore, this project aims to evaluate the target model using the principles of the Cost-aware Feasible Attacks (CaFA) framework. By simulating adversaries under varying “cost budgets,” this project investigates the fundamental trade-off between an attack's mathematical efficacy (its ability to force a misdiagnosis) and its clinical plausibility (its ability to remain undetected by human clinicians).

3. Objective of the Study

The primary objective of this study is to classify patients as healthy or exhibiting heart disease using a Multi-Layer Perceptron (MLP), ensuring strict methodological hygiene (zero data leakage) through a validated Scikit-Learn pipeline.

The secondary, and focal, objective is to perform a comprehensive socio-technical security audit on a Multi-Layer Perceptron (MLP) trained to classify the presence of heart disease. Specifically, it aims to evaluate the model's adversarial robustness using the **Cost-aware Feasible Attacks (CaFA)** framework.

The study analyzes how strategically distributed, low-budget adversarial noise—modifying critical clinical features such as resting blood pressure (trestbps), serum cholesterol (chol), and maximum heart rate (thalach)—can degrade the model's internal confidence (ROC-AUC) and alter its hard diagnostic decision boundary (F1-Score). A key objective is proving whether an attacker can successfully force a False Negative (misdiagnosing a sick patient as healthy) while strictly enforcing structural database constraints (e.g., integer rounding) and remaining clinically imperceptible to anomaly detectors.

4. Dataset Description

UCI Heart Disease Dataset ([kaggle](#))

- **Number of samples:** 297 patients recorded
- **Type (tabular clinical data):** The dataset contains 13 demographic and clinical features that are commonly used for cardiovascular assessment. These include patients' age, sex, as well as physiological measurements like blood pressure, cholesterol level etc etc.
- **Available labels:** binary diagnosis classes representing presence or absence of heart disease. (Healthy=0 or Ill=1)
- **Issues (Class Imbalance and Noise):** The dataset is relatively balanced between healthy and heart disease cases, reducing the risk of classification bias. However, the dataset contains 297 patients' records, which is a relatively small sample size. The primary challenge is extreme data scarcity ($N=297$). As demonstrated by Varoquaux (2018), small sample sizes in medical ML lead to large confidence bounds and high variance in performance estimates. To mitigate this instability and strictly prevent data leakage, nested cross-validation pipelines were implemented to completely isolate the test set and avoid optimistic bias.

5. Data Preprocessing

The initial data preparation focused on formatting the clinical data for supervised learning while strictly avoiding any transformation which could lead to data leakage.

Class Imbalance check: The full dataset of 297 instances is relatively well balanced, which is critical for preventing classification bias. This balance was preserved during splitting, yielding a stratified test set of 32 healthy and 28 sick patients.

Classification Report:				
	precision	recall	f1-score	support
Healthy (0)	0.81	0.91	0.85	32
Sick (1)	0.88	0.75	0.81	28

Data Partitioning for Supervised Learning In alignment with the core principles of supervised classification, the dataset was explicitly separated into two distinct structures prior to training. The 'condition' variable was extracted to serve as the ground-truth target array (**y**), representing the binary diagnostic label. The remaining 13 physiological attributes were isolated to form the design matrix (**X**). This formal separation is the foundational step for training the Multi-Layer Perceptron (MLP); it allows the network to process the input features (**X**), generate a predictive diagnostic score, and iteratively adjust its internal weights by evaluating its predictions against the true clinical labels (**y**).

Deferred Normalization: Feature normalization was not applied before splitting the dataset because doing so would introduce data leakage. Instead, normalization was performed inside the machine learning pipeline, ensuring that scaling parameters were learned only from the training data and then applied to the test data.

6. Avoiding Data Leakage

To ensure the test set did not contaminate the training process is a core requirement for reliable medical ML evaluation. The evaluation for this project is structured using a strict **Nested Hold-Out Validation scheme**:

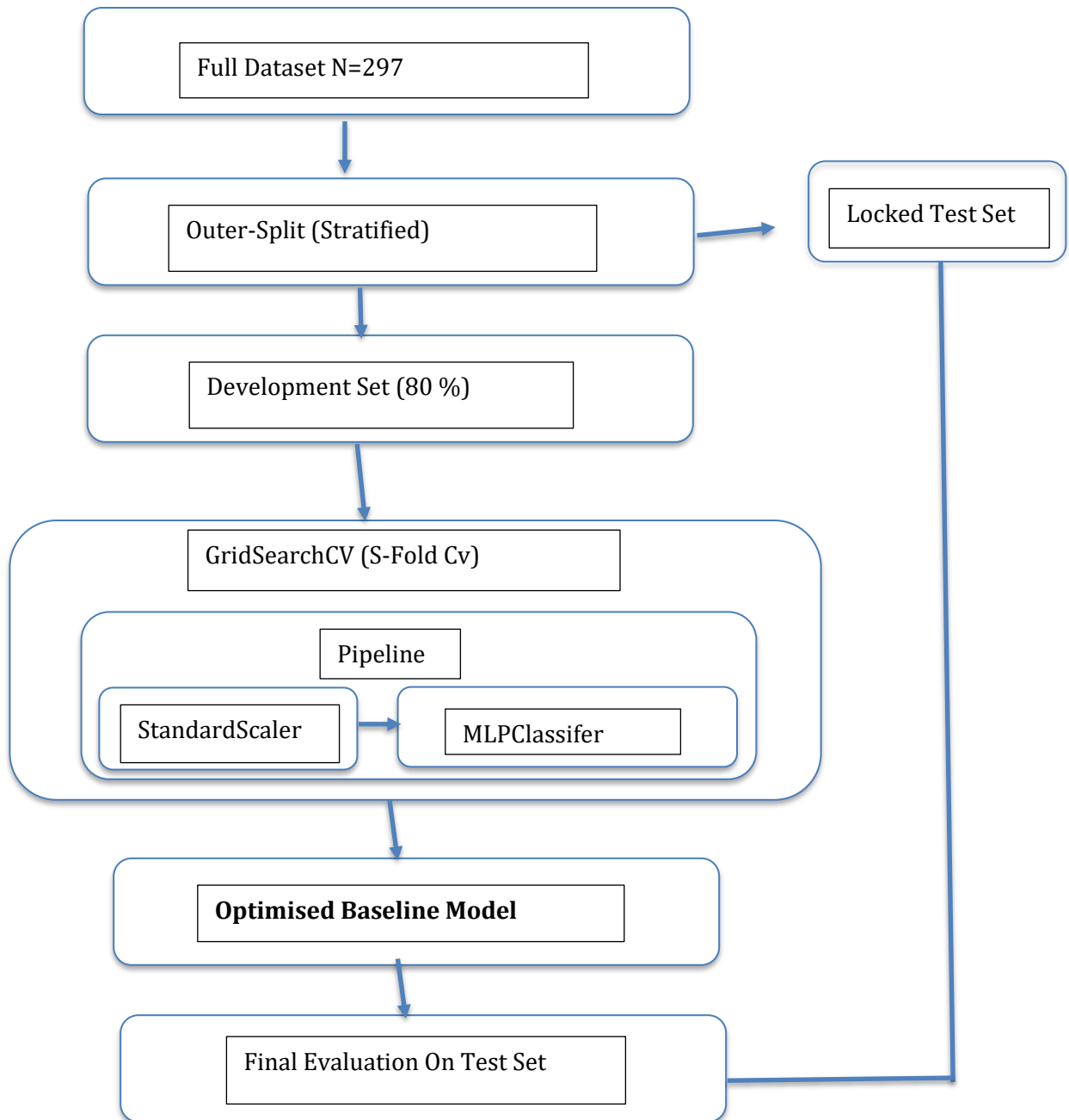
1. **Outer Split:** The data was split into an 80% development set and a 20% test set (N=60). To ensure strict reproducibility and preserve clinical class proportions, this split was performed using a fixed random seed and class stratification. The test set was then strictly isolated.
2. **The Scikit-Learn Pipeline:** To prevent preprocessing leakage, the **StandardScaler** and **MLPClassifier** were encapsulated together within a **scikit-learn Pipeline**.
3. **Inner Split:** During hyperparameter tuning, **GridSearchCV** evaluated the pipeline using 5-fold cross-validation solely on the 80% development set. Because the **StandardScaler** was inside the pipeline, normalization statistics were calculated exclusively on the training folds, guaranteeing zero contamination of the validation data..

7. Machine Learning Pipeline

I have divided the methodology into two phases: training the clean baseline and executing the adversarial audit.

7.1. The Clean Predictive Pipeline

The classification model is a Multi-Layer Perceptron (**MLPClassifier**). To avoid the **ConvergenceWarning** and ensure the weights were fully optimized, the network was trained with `max_iter=1000`. Hyperparameter tuning (optimizing hidden layer architectures) was conducted using **GridSearchCV**. The final model's generalization error was evaluated on the unseen 20% outer test set using the F1-Score and the Area Under the ROC Curve (AUC) .



7.2. The Socio-Technical Security Audit (Threat Model)

To evaluate the model's adversarial robustness, the locked pipeline was subjected to a simulated evasion attack.

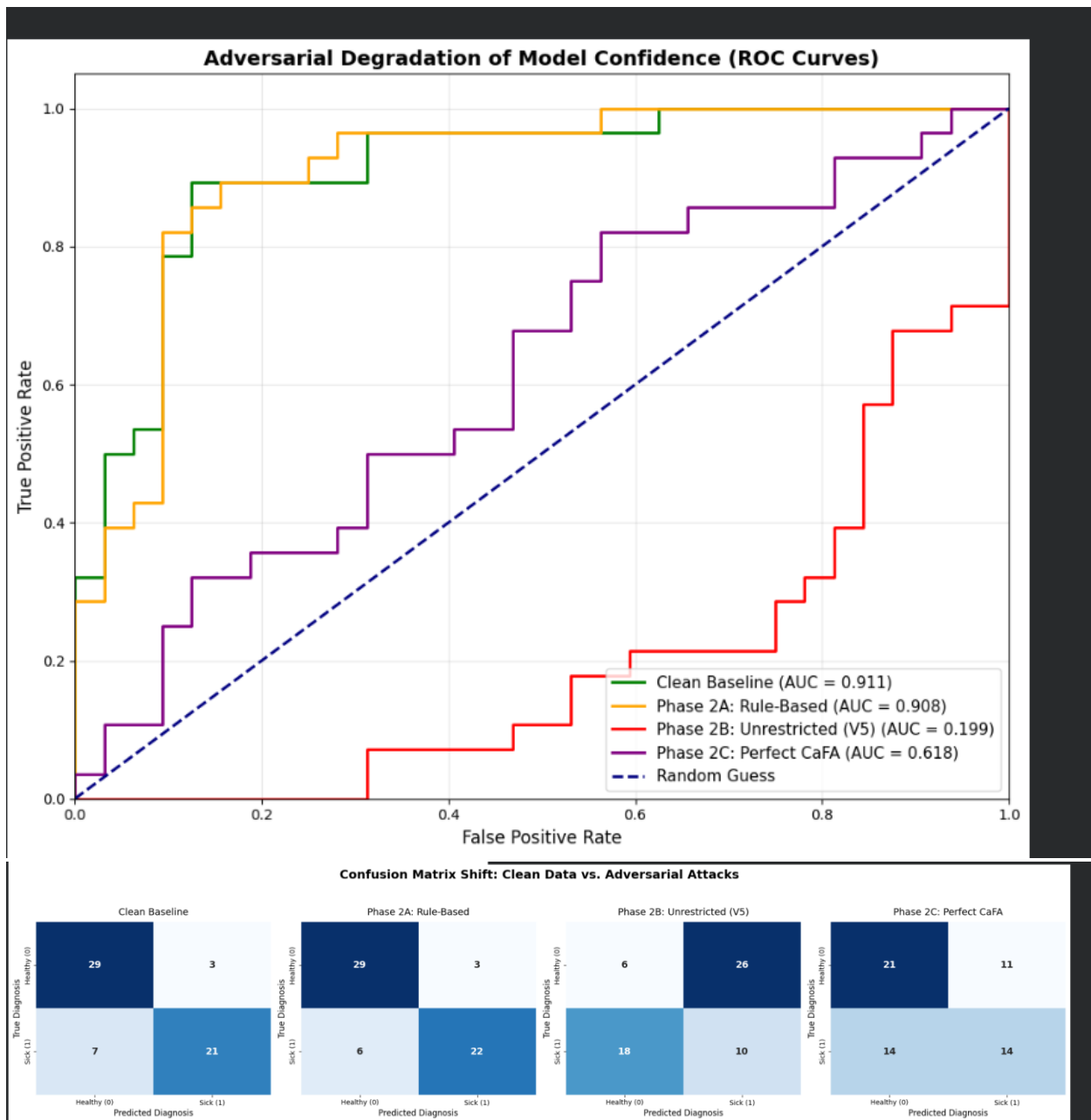
- **The Heuristic Algorithm:** While inspired by the cost-aware feasibility principles of the CaFa framework (Ben-Tov et al., 2024), the attack implemented in this study is a custom greedy coordinate-search heuristic. At each step, the algorithm tests specific feature modifications, retains the perturbation that drops the true-class probability the most, and enforces strict integer rounding to satisfy tabular domain constraints.
- **The Threat Model (Score-Based White-Box):** It is important to note that this audit assumes a fully white-box, label-informed attacker. The greedy search is granted oracle access to both the model's internal predict_proba outputs and the ground-truth labels at every step. While this represents a worst-case scenario that a real-world patient would likely not possess, it establishes a rigorous upper bound on the model's vulnerability to targeted feature spoofing.
- **Perturbation Budget:** To measure the trade-off between mathematical robustness and clinical plausibility, the attacker's capabilities were iteratively expanded: from a 20% manual modification cap, to an unconstrained brute-force attack, and finally to a strictly constrained 25% distributed attack across five features.

8. Results

Present key results (preferably in a table or with figures):

- ROC curve, confusion matrix, performance scores
- Comparison of different models, if applicable

Evaluation Phase	ROC-AUC	F1-Score (Sick)	False Negatives (Missed Diagnoses)	False Positives (False Alarms)
Clean Baseline	0.9107	0.81	7	3
Phase 2A: Naïve Approach	0.9085	0.83	6	3
Phase 2B: Unrestricted	0.1987	0.31	18	26
Phase 2C: Expanded CaFa	0.6183	0.53	14	11



9. Discussion

Interpretation of Results: The results empirically demonstrate the "Robustness vs. Plausibility Trade-Off" in tabular medical AI. **Phase 2A** proved that simple, manual percentage shifts are insufficient to break a well-trained MLP. **Phase 2B** proved that while an unconstrained attacker can easily bypass the model (dropping the F1-Score to 0.3125), the resulting 80% shifts violate biological constraints and would be flagged by hospital anomaly systems.

The most significant finding is **Phase 2C** (Inspiration taken from the help CaFA: Cost-aware, Feasible Attacks). By distributing a stealthy 25% perturbation budget across five features and enforcing integer rounding, the attack successfully bypassed the multidimensional decision

boundary while remaining clinically plausible. This dropped the ROC-AUC to 0.6183 and the F1-Score to 0.5283, proving that the network relies on fragile decision boundaries that can be covertly exploited.

Limitations: While the classes are relatively well-balanced, the primary limitation of this audit is the extreme data scarcity (N=297). As demonstrated by Varoquaux (2018), small sample sizes in medical machine learning lead to large confidence bounds and high variance in performance estimates. Evaluating models on a limited test set of 60 patients means that shifting just a few predictions significantly alters the performance metrics. To mitigate this instability, nested cross-validation pipelines were implemented to isolate the test set and avoid optimistic bias.

10. Conclusions and Future Work

Short Conclusions: In this study, I constructed a rigorously validated Scikit-Learn Pipeline, achieving a highly confident baseline (ROC-AUC: 0.9107, Positive F1-Score: 0.81). Through a multi-phase adversarial audit, I empirically demonstrated the Robustness vs. Plausibility Trade-Off in tabular medical AI. While unconstrained attacks easily degraded the model, they produced biologically impossible patient profiles. However, by executing an Expanded Cost-aware Feasible Attack (CaFA)—distributing a strictly constrained 25% perturbation budget across five features while enforcing clinical integer constraints—the adversary successfully bypassed the decision boundary. This clinically plausible attack covertly collapsed the ROC-AUC to 0.6183 and the positive F1-Score to 0.53. Ultimately, this proves that while medical MLPs resist isolated noise, they remain critically vulnerable to distributed, low-budget feature spoofing that is entirely imperceptible to clinicians.

Future work: While this study utilized a heuristic white-box greedy search to preserve the integrity of the scikit-learn pipeline and prevent data leakage, future work should explore integrating gradient-based constrained optimizations (such as TabPGD) to calculate exact adversarial perturbations.

11. Ethics and Data Privacy

Describe how ethical and privacy considerations were addressed in the project.

- Confirm that the dataset used is publicly available and permitted for academic or research use.
- State any licenses or usage restrictions attached to the data or to any third-party code used.

If applicable, include a reference to the data source's license or terms of use.

The dataset utilized in this study (Cleveland Heart Disease Dataset) is publicly available via the UCI Machine Learning Repository and Kaggle. It is explicitly permitted for academic and research use. All patient records in this dataset have been fully anonymized and de-identified prior to publication, ensuring that no Personally Identifiable Information (PII) is exposed. The use of this data complies with standard ethical research guidelines and does not violate HIPAA or GDPR privacy regulations.

Algorithmic Fairness and Trustworthy AI: However, the ethical deployment of medical machine learning extends far beyond data anonymization; it requires the active development of

Trustworthy AI. Adversarial vulnerabilities in clinical models do not merely degrade overall

diagnostic accuracy—they can actively widen demographic biases and cause algorithmic fairness metrics (such as Statistical Parity Difference) to collapse under stress (*Muhammad et al. (2026)*). Therefore, conducting a socio-technical security audit to ensure adversarial robustness is not just a technical requirement, but a fundamental ethical obligation to prevent algorithmic discrimination and ensure equitable patient care. Furthermore, to uphold scientific trust and computational reproducibility, all methodological constraints, strict data leakage protections, and random seeds have been fully documented to ensure transparent and verifiable results.

12. Code and Reproducibility

- Name of the Python notebook (.ipynb) developed in Google Colab
- Any notes for reproducing the results (dependencies, instructions)

- **Python Notebook:** Heart_Disease_Security_Audit_ZainabZahara.ipynb.
- **Notes for Reproducibility:** To adhere to the principles of "Preproducibility," a fixed random seed (random_state=42) has been explicitly enforced across all functions.
- **Libraries:** It uses standard Python libraries (pandas, scikit-learn, matplotlib).

13. References

Include all relevant references cited in the report.

You can use any standard citation format (e.g., APA, IEEE) consistently throughout the document.

1. Ben-Tov, M., et al. (2024). "CaFA: Cost-aware, Feasible Attacks With Database Constraints Against Neural Tabular Classifiers."
2. Dyrmishi, S., et al. (2025). "Insights on Adversarial Attacks for Tabular Machine Learning via a Systematic Literature Review."
3. Muhammad, et al. (2026). "Trustworthy AI for medical decisions: Adversarially robust and fair machine learning prediction for Parkinson's disease."
4. Varoquaux, G. (2018). "Cross-validation failure: Small sample sizes lead to large error bars." *NeuroImage*, 180:68-77.